

Statistics is _____.

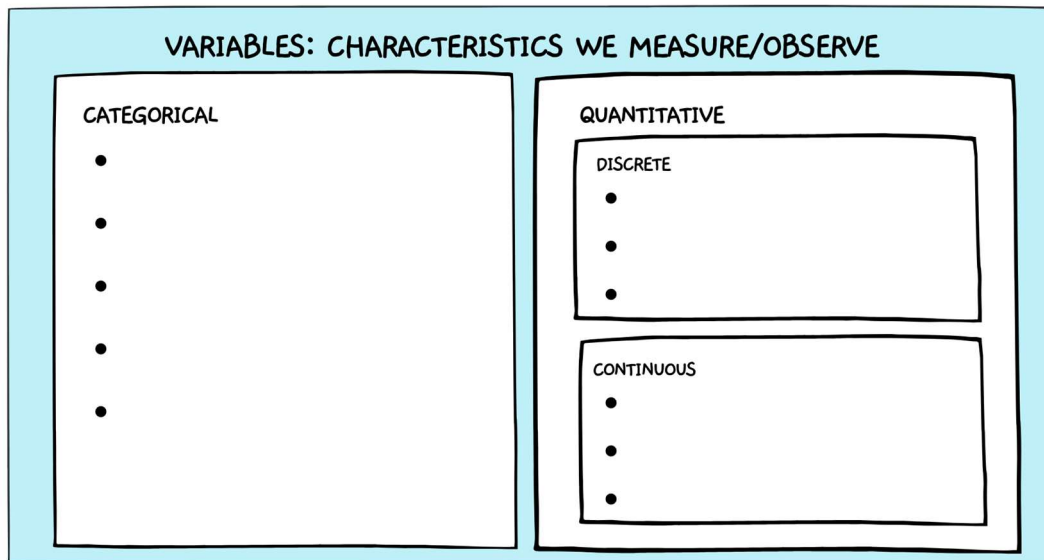
Data contains information about a group of *individuals*. The information is organized using *variables*.

Individuals are objects described by a set of data. Individuals may be people but may be animals or inanimate objects.

Variables are characteristics of individuals. A variable may take on different values of different variables. Variables can be split into two types: categorical or quantitative.

Categorical variables place individuals into specific groups.

Quantitative variables takes on numerical values for which it makes sense to do arithmetic operations like adding and averaging. Quantitative variables fall into two categories: discrete and continuous.



****Be careful, just because it is a number DOES NOT automatically make it quantitative. Can you think of an example of a number that is categorical?**

Discrete variables are numerical values where counting makes sense; in other words, decimals would not be an appropriate way to record the data.

Continuous variables are numerical values where decimals are appropriate; it usually involves some form of measuring.

****Note:** The difference between discrete and continuous isn't always clear: For instance, is age continuous or discrete? You use whole numbers to describe it, but it can be measured with a decimal.

****If it is not obvious if it is discrete or continuous, like our age example, it depends on how you use the data. For age, we rarely say "I'm 16.4 years old", so we treat age like a discrete variable.**

Representing Categorical Data: Tables

One of the easiest ways to display categorical data is with a table. Here is a list of the first 10 US presidents, their political party and their state of birth.

George Washington	Federalist	Virginia
John Adams	Federalist	Massachusetts
Thomas Jefferson	Democratic-Republican	Virginia
James Madison	Democratic-Republican	Virginia
James Monroe	Democratic-Republican	Virginia
John Quincy Adams	Democratic-Republican	Massachusetts
Andrew Jackson	Democrat	South Carolina
Martin Van Buren	Democrat	New York
William H. Harrison	Whig	Virginia
John Tyler	Whig	Virginia

Let's start with a few **one-way tables**:

<i>Category</i>	<i>Count</i>	<i>Relative Count</i>	<i>Category</i>	<i>Frequency</i>	<i>Relative Frequency</i>
Federalist			Virginia		
Democratic-Republican			New York		
Democrat			South Carolina		
Whig			Massachusetts		

If we wanted to display two categorical variables at a time, we would make a **two way table**:

		Political Party				Total
		Federalist	Democratic-Republican	Democrat	Whig	
State of Birth	Virginia					
	New York					
	South Carolina					
	Massachusetts					
Total						

To better visualize the data, we can also make various graphs from our data. We want to visualize the graphs to get a better idea of the *distribution*.

The **distribution** of a variable tells us what values the variable takes and how often it takes these values.

Bar Graphs

Bar Graphs have the following important characteristics:

- Label each axis clearly (this goes for all graphs).
- The x-axis will contain the categorical variable and the y-axis will display the counts (or percentages).
- Each category has its own bar and the bars CANNOT touch.
- Order is not important when creating the x-axis.

Create a bar graph showing the distribution of the four political parties of the first 10 US presidents.

Questions we can answer from a bar graph:

- 1) Of the first 10 presidents, what political party was the most affiliated with?
- 2) What political party was the least affiliated with?
- 3) What are the individuals in this data set? What are the variables of the data set?

*We can also use Pie Graphs for categorical data, but those are not covered on the AP exam.

*To graph two way tables, we will explore segmented bars graphs and mosaic plots in Unit 2.